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Comparing Penile Problems in Circumcised vs. Uncircumcised Boys: Insights From a Large Commercial Claims Database With a Focus on Provider Type Performing Circumcision

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Abstract

Objectives: To compare penile problems in circumcised relative to uncircumcised boys, and to determine which providers performing the circumcision have fewer post-circumcision problems.

Methods: CPT codes in the 2011-2020 MarketScan database were used to identify boys who had a circumcision. Uncircumcised control subjects of the same age, state of residence, and insurance type were selected. The primary outcome was a penile problem, defined as penis-specific infection, inflammation, and urethral stricture/stenosis, among others. The secondary outcomes were procedure-related complications limited to 28 days after circumcision, and whether post-circumcision problems varied by the clinician performing the procedure. ICD-9/10 diagnostic codes were used to identify these problems.

Results: We identified ~850,000 cases and ~850,000 matched controls. Overall, the rate of penile problems within the first five years of life was 1.7% in circumcised boys versus 0.5% in uncircumcised boys ($p < 0.05$). Multivariable regression models showed that the risk of penile problems was 2.9-fold higher among circumcised compared to uncircumcised males (95%CI [2.8-3], $p < 0.001$). Compared to males circumcised by pediatricians, those circumcised by surgeons had 2.1-fold higher penile problems in the year after circumcision (95% CI [2-2.3], $p < 0.001$). Procedure-related complications within 28 days of circumcision were infrequent (0.5%), with the most common being penile edema (0.2%).

Conclusions: Penile problems are very infrequent in boys in the first five years of life. However, when they occur, they are 3x more likely to occur in circumcised boys relative to uncircumcised boys. Penile problems are more likely to occur in boys circumcised by surgeons.

Levels of evidence: Level II.